Induction Heads and Model Collapse: How the Real-to-Synthetic Data Ratio Shapes In-Context Learning Circuits

Lily de Melo 2545080
University of the Witwatersrand, 1 Jan Smuts Ave,
Braamfontein, Johannesburg, 2000, South Africa

Gopolang Mohlakola 2680747
University of the Witwatersrand, 1 Jan Smuts Ave,
Braamfontein, Johannesburg, 2000, South Africa

Tawananyasha Kadango 2588206
University of the Witwatersrand, 1 Jan Smuts Ave,
Braamfontein, Johannesburg, 2000, South Africa

Abstract

Model collapse is the decline of models retrained on data generated by earlier models. To date, collapse has been measured through model outputs, which would understate the decline if the circuits behind those outputs degraded first. This study therefore traces the induction circuit behind in-context learning (ICL) in two-layer transformers retrained on data that the previous model partly or wholly relabels, or adds to the real data. A tuned model reproduces its own labels almost exactly, so that accuracy holds under every real-to-synthetic ratio. Once errors are forced through a raised sampling temperature, however, mechanistic changes precede accuracy loss: the logit margin halves and the circuit spreads while accuracy barely changes, and the margin predicts the next generation's error rate, so that errors compound. Retaining real data limits how far accuracy falls, and accumulation delays that fall, but neither stops the margin's erosion. Overall, real data protect a model's behaviour for longer than its circuit, so that collapse may show in the circuit first.

Keywords: model collapse; induction heads; mechanistic interpretability

Introduction

Recursively training large language models (LLMs) on data generated by other models causes *model collapse*, a loss of information about the true data distribution (Shumailov et al., 2024), which retaining real data appears to avoid (Gerstgrasser et al., 2024). Both findings, however, rest on model outputs, which would understate collapse if the underlying circuits degrade first.

In-context learning (ICL) lets models adapt to new tasks from prompt examples without parameter updates (Olsson et al., 2022; Singh et al., 2024). In the task of Singh et al. (2024), each sequence pairs symbols with arbitrary labels, so that the label of a query symbol can only be found by matching the query to the same symbol in the context and copying the label that follows. An induction circuit of known components performs this match and copy, so that damage can be traced to specific heads rather than inferred from accuracy. This study asks how the share of real data shapes this circuit.

Background

Induction heads A previous-token head in the first layer copies each symbol into the following label, and an induction head in the second layer matches the query against these copies and copies the matching label. Chen et al. (2026) trace the formation of these heads to specific training data, suggesting that the data mix can reshape the circuit.

Collapse and accumulation Collapse arises mainly from sampling error, which loses the tails of the distribution first and compounds across generations (Shumailov et al., 2024). Replacing real data makes test error grow each iteration, whereas adding synthetic data to the real data keeps the error bounded (Gerstgrasser et al., 2024), a contrast tested here in the circuit.

Methods

This study uses the setup of Singh et al. (2024), in which each sequence holds two symbol–label pairs and a query, with Omniglot exemplars (Lake et al., 2015) encoded by a frozen ResNet18 (He et al., 2016) and a set of five labels reassigned per sequence. A causal two-layer attention-only transformer with eight heads per layer trains on 50 classes for one million sequences, at learning rates of 10^{-4} (base) and 3×10^{-4} (extended) tuned on 1,473 validation classes, and is tested on 100 unseen classes. The base model predicts the query's label, and the extended model also predicts a next symbol and that symbol's label, so that three tokens per sequence become synthetic (Figure S1). The next symbol comes from the context in real data but from all 50 training classes in synthetic data.

Real-to-synthetic ratio Each run trains on a pool of $N = 20{,}000$ sequences for up to 20 generations, the first generation on real data only, with a fixed training length and three seeds per condition. The previous model then regenerates the targets of 0%, 50%, 90%, or 100% of the pool from their real prompts, where 90% matches the 10% of real data kept by Shumailov et al. (2024). Accumulation instead adds N synthetic sequences per generation (Gerstgrasser et al., 2024). Synthetic tokens are sampled from the softmax of the logits divided by a temperature τ, so that $\tau > 1$ flattens the distribution and makes wrong labels likelier. As the tuned model makes almost no errors at $\tau = 1$, errors are forced, or introduced deliberately, at $\tau = 3$ and 5.

Circuit analysis The model's margin, the logit of the correct label minus that of the other context label, measures how confidently the model answers. As the next generation's labels are sampled at τ, the mean margin D_g of generation g also predicts their error rate: taking D_g as the correct label's margin over each of the four wrong labels, sampling makes a share

$$\hat{\varepsilon}_{g+1} = \frac{4\,e^{-D_g/\tau}}{1+4\,e^{-D_g/\tau}} \tag{1}$$

of these labels wrong. Automatic Circuit DisCovery (ACDC) (Conmy et al., 2023) compares each sequence with a copy in which the two context symbols trade places, so that matching the query leads to the wrong label. ACDC keeps only edges whose corruption changes the output by more than $\theta = 0.001$. In the induction head, the match is the attention on the label after the matching symbol, and the copy is measured by the head's direct effect, the share of the model's margin that comes from this head's output alone.

Results

No errors, no collapse At generation 0, both models solve the task by ICL: the base model reaches a mean test accuracy of 0.98 and the extended model 0.96, with validation accuracies of 0.99 and 0.97 respectively. Without forced errors, the real-to-synthetic ratio barely matters, as at $\tau = 1$ only 0.01% of regenerated labels are wrong after 20 generations, so that every replacement level yields almost the same model (Table 1).

Table 1: Generation 19 of every 20-generation run, mean ± SD over three seeds; Real is the share of real data in the pool. †Margin below 1, where the circuit size is less reliable.

		Base model			Extended	
Condition	Real	Acc.	Margin	Edges	Margin	Edges
No forced errors ($\tau = 1$)						
Real only	100%	0.98 ±0.01	17.5 ±2.2	23 ±5	15.7 ±0.3	42 ±7
100% replaced	0%	0.98 ±0.00	12.5 ±0.3	35 ±7	12.8 ±0.9	92 ±36
Accumulate	5%	0.99 ±0.00	9.8 ±1.0	20 ±3	6.3 ±0.1	34 ±5
Forced errors ($\tau = 3$)						
50% replaced	50%	0.79 ±0.01	2.6 ±0.1	194 ±11	3.8 ±0.3	89 ±19
90% replaced	10%	0.33 ±0.01	0.4 ±0.1	207 ±3†	0.1 ±0.0	180 ±4†
100% replaced	0%	0.19 ±0.02	0.1 ±0.0	207 ±7†	−0.1 ±0.1	163 ±7†
Accumulate	5%	0.92 ±0.02	0.7 ±0.0	17 ±5†	0.6 ±0.0	11 ±2†

Confidence and circuit change before accuracy Once errors are forced at $\tau = 3$, the model's confidence and circuit change before the model starts answering wrongly. After one generation of full replacement, the margin has nearly halved, from 18.4 to 10.1, and the circuit has more than tripled, from 22 to 74 edges on average (17 and 55 in Figure 1), yet accuracy is still 0.95 or above. Only by generation 3, with the circuit spread across 211 edges, does accuracy fall, to 0.66 (Figure S3). The growth does not depend on ACDC's threshold, as the number of edges needed to recover 80% of the margin rises in the same way, from 33 to 120, while real-only retraining stays near 30.

The match holds, the copy weakens Inside the induction head, finding the match and copying the label fail at different times (Figure 2). Under full replacement, the head still attends to the matching label, keeping 88% of its generation 0 attention by generation 2, whereas its direct effect keeps only 41%, so that the head still finds the answer but copies the label far more weakly. The head's attention collapses only at generation 4, after accuracy has fallen. Under accumulation, attention stays at 0.89 or above for 15 generations while the direct effect falls by 86%, so that real data preserve the head's match but not its copy.

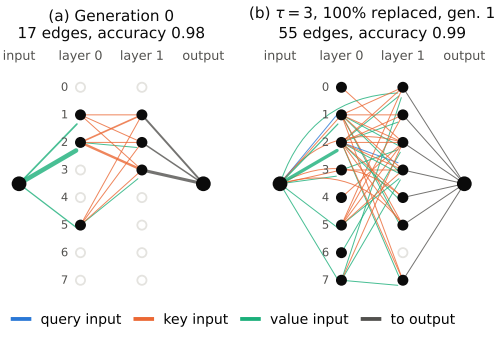


Figure 1: ACDC circuits of the base model (one seed; heads named A*layer.head*): previous-token head A0.2 feeds induction head A1.3 (a), and one generation at $\tau = 3$ triples the circuit (b).

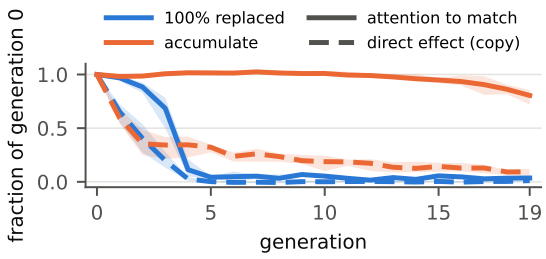


Figure 2: Induction-head match and direct effect at $\tau = 3$, mean of three seeds with the range shaded.

The margin sets the next generation's errors Equation 1 predicts the share of wrong labels in each new pool to within 2.1 percentage points on average across both models, temperatures, and conditions (correlation $r = 0.994$). A falling margin therefore writes more errors into the next pool, lowering the next margin, a loop that only real labels can slow.

The real share sets how far collapse goes Over 20 generations at $\tau = 3$, the more real data a pool keeps, the higher accuracy settles: 0.79 when half the pool stays real, 0.33 with a tenth, and 0.19 with none (Table 1, Figure S2). Yet even half leaves the circuit spread across 194 edges, so the real share bounds accuracy but not the circuit. Accumulation holds accuracy at 0.96 or above for 15 generations while the margin falls by 96%, and accuracy declines only once the margin is below 1. The extended model's query accuracy settles at 0.84, 0.34, and 0.20, with the same split in a second output. Under accumulation, its query accuracy stays at 0.93, but the raised temperature scatters its next symbols. At $\tau = 1$, its symbol head puts 13% of its probability on context symbols, against 98.6% under real-only retraining.

Discussion

The real-to-synthetic ratio shaped the induction circuit mainly once synthetic data carried errors. The share of real data set how far accuracy and the margin fell, as only real labels dilute the errors that a falling margin writes into each new pool, extending Shumailov et al. (2024) with the internal quantity that sets the sampling error. At every real share, however, the circuit changed before accuracy did: the margin fell, the circuit first spread, and within the induction head the copy weakened while the match held, suggesting that the copying operation of Singh et al. (2024) is the more fragile. Accumulation protected accuracy but not the margin or the copy, consistent with the bounded error of Gerstgrasser et al. (2024). These conclusions are bounded by forcing errors through temperature rather than a weaker model, and by a training length fixed while the accumulated pool grows 20-fold. A version that restricts the extended model's generated next symbols to the context, as in the real data, has been run, but its analysis is left for further work. Overall, real data protected a model's behaviour for longer than its confidence, so that the margin and the circuit could flag collapse before accuracy does.

Contribution Statements

We, Lily de Melo, Tawananyasha Kadango, and Gopolang Mohlakola, worked together fairly on this project.

References

Chen, J., Luo, Y., & Pan, L. (2026). Mechanistic data attribution: Tracing the training origins of interpretable LLM units. In *International conference on machine learning* (Vol. 306). PMLR.

Conmy, A., Mavor-Parker, A., Lynch, A., Heimersheim, S., & Garriga-Alonso, A. (2023). Towards automated circuit discovery for mechanistic interpretability. In *Advances in neural information processing systems* (Vol. 36, pp. 16318–16352).

Gerstgrasser, M., Schaeffer, R., Dey, A., Rafailov, R., Pai, D., Sleight, H., ... Koyejo, S. (2024). Is model collapse inevitable? Breaking the curse of recursion by accumulating real and synthetic data. *arXiv preprint arXiv:2404.01413*.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the ieee conference on computer vision and pattern recognition* (pp. 770–778).

Lake, B. M., Salakhutdinov, R., & Tenenbaum, J. B. (2015). Human-level concept learning through probabilistic program induction. *Science*, *350*(6266), 1332–1338.

Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., ... others (2022). In-context learning and induction heads. *arXiv preprint arXiv:2209.11895*.

Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., & Gal, Y. (2024). AI models collapse when trained on recursively generated data. *Nature*, *631*(8022), 755–759. doi: 10.1038/s41586-024-07566-y

Singh, A. K., Moskovitz, T., Hill, F., Chan, S. C. Y., & Saxe, A. M. (2024). What needs to go right for an induction head? A mechanistic study of in-context learning circuits and their formation. In *International conference on machine learning* (Vol. 235, pp. 45637–45662). PMLR.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract states the three main findings, and the introduction the question they answer, namely that, once synthetic data carry errors, the model's confidence and induction circuit change before its accuracy does, that each generation's margin predicts the error rate of the next generation's data, and that retaining real data limits how far accuracy falls while accumulation delays its decline without stopping the margin's erosion. Each claim is supported in the Results (Table 1, Figures 1 and 2, and Equation 1), and the abstract states that errors were forced through temperature, which bounds their scope.

   Guidelines:

   - The answer NA means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Discussion states that errors were forced through temperature rather than a weaker model, that the training length was fixed while the accumulated pool grew 20-fold, and that a version restricting the extended model's next symbols to the context has been run but not yet analysed. The Results measure how far later extended models learn off-context next symbols, and Table 1 marks circuit sizes as less reliable once the margin falls below 1. Because of the two-page limit, the paper does not state that this study is also limited to one two-layer attention-only model and one synthetic task.

   Guidelines:

   - The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [NA]

Justification: This study contains no theorems. Equation 1 follows from the softmax under the approximation stated with it, that the margin over the other context label holds over each of the four wrong labels, and is tested empirically rather than proven.

Guidelines:

- The answer NA means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The Methods specify the settings that the main claims depend on, namely the task, architecture, class splits, pool size, real-to-synthetic ratios, sampling temperatures, tuned learning rates, training length, and Automatic Circuit DisCovery (ACDC) threshold. The Adam optimiser, a batch size of 32, a model width of 64, and the initialisation and training seed pairs (5, 0), (15, 10), and (25, 20) were left out of the paper because of the two-page limit, and are fixed in the released run scripts.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The code, the preprocessed Omniglot features, and the scripts for every experiment and figure are available at `https://github.com/froggomelo/nlp-project-1`, whose README gives the environment and the command for each experiment.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://nips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so "No" is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://nips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental Setting/Details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: The Methods state the class splits, namely 50 Omniglot training classes, 1,473 validation classes on which the learning rate and training length were tuned, and 100 unseen test classes, so that test accuracy reflects ICL alone. The Adam optimiser, the batch size of 32, and the model width of 64 are not needed to understand the results, and were left out of the paper because of the two-page limit but are fixed in the released run scripts.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment Statistical Significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 1 reports the mean $\pm$ the standard deviation over three seeds of the initialisation and training data. Figures 2, S2, and S3 show the three-seed mean with the range across seeds shaded, the text reports means, and the fit of Equation 1 is reported as a correlation and an average absolute gap across every generation and condition.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments Compute Resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

   Justification: The compute was left out of the paper because of the two-page limit. Every experiment ran on the CPU of a single desktop computer (AMD Ryzen 7 7800X3D, 32 GB of RAM), where training on one million sequences took about 2.2 minutes. The full run script trains 1,128 models over three seeds, namely 840 for Table 1 (14 runs of 20 generations) and 288 for Equation 1 (16 runs of 6 generations), or roughly 41 hours of training before tuning and circuit discovery. Development runs not reported here used compute of a similar scale.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code Of Ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: This study uses a public dataset of handwritten characters and small models trained for analysis. It involves no human subjects or personal data and releases no model with a risk of misuse.

   Guidelines:

   - The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader Impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [NA]

    Justification: This study is foundational research on a small model trained on a synthetic task and has no direct path to harmful applications. Its one implied application, using the margin or circuit to flag collapse before accuracy falls, is a positive use considered in the Discussion.

Guidelines:

- The answer NA means that there is no societal impact of the work performed.
- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

    Answer: [NA]

    Justification: The released models are two-layer transformers trained on Omniglot features and therefore pose no risk of misuse.

    Guidelines:

    - The answer NA means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: Omniglot (Lake et al., 2015) is used under the MIT License, and the ResNet18 ImageNet weights (He et al., 2016) are taken from torchvision under the BSD-3-Clause License. Circuit discovery uses ACDC (Conmy et al., 2023) through the auto-circuit (version 1.0.1) and TransformerLens (version 2.9.0) libraries, both under the MIT License. The training code extends the released codebase of Singh et al. (2024), which is credited in the repository.

    Guidelines:

    - The answer NA means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.

- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The released code is documented in the repository's README and in the files of `docs/`, which describe each experiment, its settings, and how it is reproduced. No data were collected from people, so that consent does not apply.

    Guidelines:

    - The answer NA means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [NA]

    Justification: This study involves neither crowdsourcing nor research with human subjects.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [NA]

    Justification: This study involves neither crowdsourcing nor research with human subjects.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

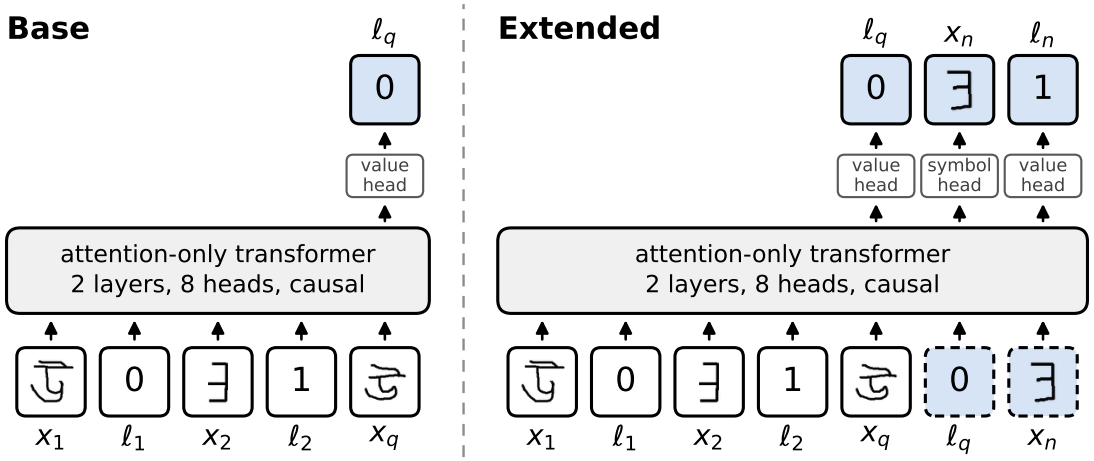


Figure S1: Base (left) and extended (right) models. The base model predicts the query's label from two symbol–label pairs and the query, and the extended model also predicts the next symbol and its label. Shaded boxes are model outputs; dashed inputs are true tokens during training and the model's own samples when it generates the next generation's data.

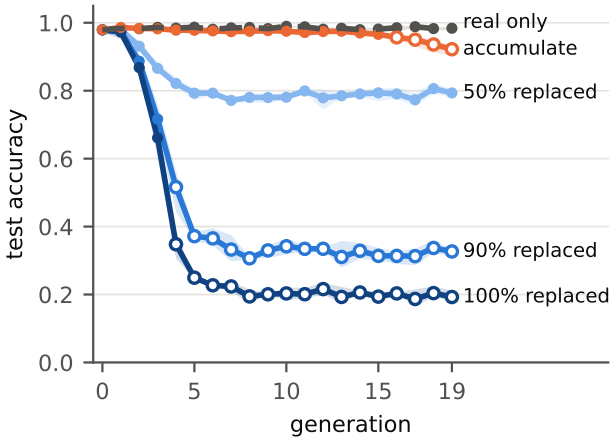


Figure S2: Test accuracy of the base model at $\tau = 3$, mean of three seeds with the range shaded. Each replacement level settles at an accuracy set by its real share, while accumulation stays with the real-only control for about 15 generations before declining. Hollow markers show a mean margin below 1.

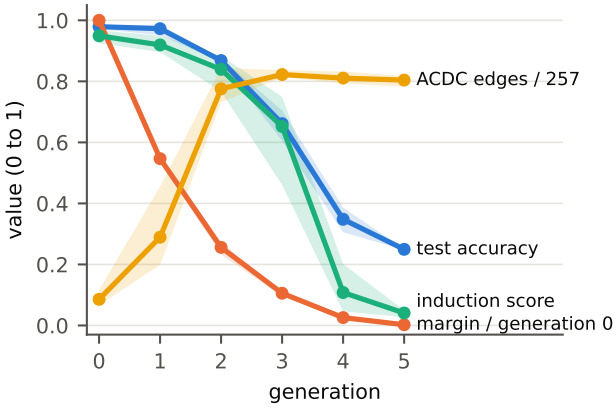


Figure S3: Order of failure of the base model under full replacement at $\tau = 3$, mean of three seeds with the range shaded. The margin halves and the circuit more than triples by generation 1, whereas accuracy and the induction score fall mainly from generation 3.

Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools [ ]

I **did** make use of generative AI tools for the following (tick all that apply):

1.	Idea Generation (research problem/design, hypothesis)	[ ]
2.	Sourcing Related Work (summarising, identifying sources)	[ ]
3.	Methods and Experiment Design (experiment setup, model tuning)	[X]
4.	Data Analysis (presentation, coding, interpretation)	[X]
5.	Theoretical Development (theorem proving, conceptual analysis)	[ ]
6.	Code Development (generating algorithms, writing scripts)	[X]
7.	Presentation (rendering graphics, formatting)	[X]
8.	Editing (grammar, readability)	[X]
9.	Writing (text generation, document structuring)	[X]
10.	Citation Formatting (structuring, organising)	[X]

If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number:

2545080 2680747 2588206

Candidate signature:

Date: 26/10/2026